

SQL VIEW — Complete Revision & Exam Notes

Exam-Oriented & Interview-Ready Guide | Gate Smashers Lecture Terminology

TARGET INTERVIEW DEFINITION

"A View is a virtual table based on a stored query. It provides a controlled representation of data from one or more base tables and is used for data access restriction, simplifying complex queries, logical data independence, and presenting user-tailored representations."

1. MEMORY TRICKS & QUICK MNEMONICS

EXAM BOOSTER

V-B-U-R-M-A MEMORY FRAMEWORK

- **V** → **Virtual Table**: No physical space for data.
- **B** → **Base Table**: Original physical storage table.
- **U** → **Updatable View**: DML affects base table.
- **R** → **Read-Only View**: Prevents modifications.
- **M** → **Materialized View**: Physically stored snapshot.
- **A** → **Advantages**: Secured and simplified access.

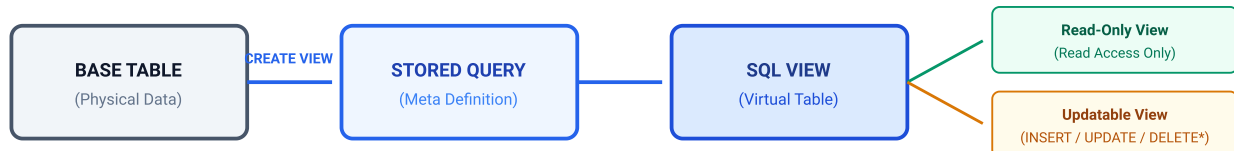
S-C-D-D ADVANTAGES FORMULA

Remember the 4 core benefits using **S-C-D-D**:

- **S** → **Security**: Restricts exposed rows/columns.
- **C** → **Complex Queries**: Simplifies JOINS & filters.
- **D** → **Data Independence**: Logical abstraction.
- **D** → **Different Views**: Multi-user access roles.

2. VIRTUAL FLOW & ARCHITECTURE

CORE CONCEPT



Syntax & Query Creation:

```
CREATE VIEW view_name AS SELECT column1, column2 FROM table_name;
```

Golden Architectural Rule:

Table → Stores physical records on disk.
View → Stores query definition in dictionary.

VIEW AND DDL VS. DML OPERATIONS

- **DDL (Data Definition Language)**: Operations such as structural alterations ('ALTER') do not apply to views in the same way as base tables. Dropping a view removes only the stored definition, leaving underlying base tables completely intact.
- **DML (Data Manipulation Language)**: Permitted 'INSERT', 'UPDATE', and 'DELETE' operations executed on an *Updatable View* directly affect data inside the base table.

3. EXPOSING SELECTED DATA & MULTI-TABLE VIEWS

ROW & COLUMN FILTERING

Columns are chosen via SELECT, while rows are filtered using WHERE clauses:

```
CREATE VIEW Delhi_Students AS SELECT ID, Name, Gender FROM Student WHERE Address = 'Delhi';
```

MULTIPLE TABLE JOINS

Combine selected columns across several base tables into a single virtual object:

```
CREATE VIEW Student_Course_VW AS SELECT S.ID, S.Name, C.CourseName FROM Student S JOIN Course C ON S.CourseID = C.ID;
```

4. BASE TABLE MUTATION REFLECTION

Base Table (Student):

ID	Name	Address
101	Steven	Delhi
102	John	Mumbai

View Output (V1):

ID	Name
101	Steven
102	John

REFLECTION MECHANICS

If we execute: UPDATE Student SET Name = 'Steven Smith' WHERE ID = 101;

Querying SELECT * FROM V1; immediately shows **Steven Smith**. Likewise, INSERT or DELETE actions on base tables dynamically mirror in the view.

5. VIEW TYPES COMPARISON

HIGH FREQUENCY TOPIC

Feature	Updatable View	Read-Only View	Materialized View
Modifications	Permitted (INSERT/UPDATE/DELETE)	Prevented	Restricted / Vendor dependent
Physical Storage	None (Virtual)	None (Virtual)	Stores Physical Snapshot
Base Table Effect	Directly modifies base rows	No modification possible	Refreshes snapshot on demand
Primary Purpose	Data entry & controlled editing	Security & read protection	Remote server / Heavy analytics



NORMAL VIEW VS. MATERIALIZED VIEW MEMORY TRICK

Normal View = Virtual (Only stores query definition, fetches live data on each call).

Materialized View = Material / Stored (Physically saves snapshot; useful when fetching data across slow networks/remote servers).

6. MULTI-USER PERSPECTIVES & REAL-LIFE EXAMPLE

REAL-WORLD APPLICATION

Different user roles require tailored perspectives over the exact same underlying database:

STUDENT ROLE

Can only access public views:

- VIEW Marks
- VIEW Courses
- VIEW Faculty Qualifications

FACULTY ROLE

Has operational rights:

- VIEW Marks
- UPLOAD Marks
- UPDATE Class Marks

ADMINISTRATOR / HOD

Has elevated rights:

- Full Base Table Access
- Faculty Salaries & Phone
- Personal Information

UNIVERSITY BASE TABLE ABSTRACTION CASE

Base Table: Faculty (ID, Name, Address, Mobile, Qualification, Salary, Personal_Details)

To prevent students from seeing private personal data or salary information, create a security view:

```
CREATE VIEW Public_Faculty_Info AS SELECT Name, Course, Qualification FROM Faculty;
```

7. IMPORTANT EXAM TRAPS (Q&A FORMAT)

MUST REWISE

• Q1: Is a View a physical table?

✗ No. It is a virtual table.

• Q2: What is actually stored for a normal view?

✓ The query definition, not an ordinary physical data copy.

• Q3: What is the original physical table called?

✓ Base Table.

• Q4: Can a view pull fields from multiple tables?

✓ Yes, using JOINS or subqueries.

• Q5: Can we select specific rows and columns?

✓ Yes (using SELECT and WHERE).

• Q6: What happens when base data changes?

✓ The view dynamically reflects corresponding updates.

• Q7: What is an updatable view?

✓ A view where permitted DML alters base table rows.

• Q8: What is a read-only view?

✓ A view configured to prevent modifications.

• Q9: What is a materialized view?

✓ A view that physically stores a snapshot on disk.

• Q10: Does dropping a view drop base data?

✗ No, only the query structure is removed.

8. 1-MINUTE FINAL REVISION TREE

```

VIEW (Virtual Table based on Stored Query) | — Base Table —————> Original physical data source | — Specific Columns/Rows —> Filtered via SELECT and WHERE | — Multiple Tables —————> Supported via SQL JOINS | — Type s: | — Updatable —————> Modifications affect Base Table | — Read-Only —————> Modifications prevented | — Materialized —————> Physically stores result snapshot | — Advantages (S-C-D-D): | — S —————> Security / Restrict Data Access | — C —————> Complex Queries Become Simple | — D —————> Data Independence | — D —————> Different Views for Different Users
    
```